# MA 102 (Mathematics II - Ordinary Differential Equations)

**Department of Mathematics, IIT Guwahati**
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**Lectures 15-16: Systems of first-order equations, Matrix form of first-order systems**

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1. Represent each of the following equations in terms of an equivalent system of first-order equations:
(a) $x'' - t^2 x' - tx = 0$, (b) $x''' = x'' - t^2(x')^2$, (c) $x'' - 3x' - 11x = \sin t$.
Further, write the derived systems in matrix form $\bar{x}' = A\bar{x} + \bar{b}$ (if possible).
**Solution:** (a) It can be written as a system as follows:

$$\frac{dx}{dt} = y,$$
$$\frac{dy}{dt} = tx + t^2 y.$$

This system has the matrix form as follows:

$$\begin{bmatrix} x \\ y \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ t & t^2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

(b) Take $\frac{dx}{dt} = y$ to get $\frac{d^2y}{dt^2} = \frac{dy}{dt} - t^2y^2$. Next, take $\frac{dy}{dt} = z$ to get the system as

$$\frac{dx}{dt} = y,$$
$$\frac{dy}{dt} = z,$$
$$\frac{dz}{dt} = z - t^2y^2.$$

It is not possible to represent this system in $\bar{x}' = A\bar{x} + \bar{b}$ form since $y$ appears nonlinearly.
(c) It can be written as a system as follows (non-homogeneous):

$$\frac{dx}{dt} = y,$$
$$\frac{dy}{dt} = 11x + 3y + \sin t.$$

This system has the matrix form as follows:

$$\begin{bmatrix} x \\ y \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ 11 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 0 \\ \sin t \end{bmatrix}.$$

2. If a particle of mass moves in the $xy$-plane, its equations of motion are

$$m\frac{d^2x}{dt^2} = f(t, x, y), \quad m\frac{d^2y}{dt^2} = g(t, x, y),$$

where $f$ and $g$ represent the $x$ and $y$ components, respectively, of the force acting on the particle. Express this system of two second-order equations by an equivalent system of four first-order equations.
**Solution:** It is easy to follow that the required system is

$$\begin{aligned} \frac{dx}{dt} &= u, \\ \frac{du}{dt} &= \frac{f(t,x,y)}{m}, \\ \frac{dy}{dt} &= v, \\ \frac{dv}{dt} &= \frac{g(t,x,y)}{m}. \end{aligned}$$

3. The vector functions $\bar{x_1} = [e^{-t}, 2e^{-t}, e^{-t}]^T$, $\bar{x_2} = [e^t, 0, e^t]^T$, $\bar{x_3} = [e^{3t}, -e^{3t}, 2e^{3t}]^T$ are solutions to the system $\bar{x}'(t) = A\bar{x}(t)$. Determine whether they form a fundamental solution set. If they do, find a fundamental matrix for the system and give a general solution.
**Solution:**
Given $\bar{x_1} = [e^{-t}, 2e^{-t}, e^{-t}]^T$, $\bar{x_2} = [e^t, 0, e^t]^T$, $\bar{x_3} = [e^{3t}, -e^{3t}, 2e^{3t}]^T$.
Now,

$$W(\bar{x_1}, \bar{x_2}, \bar{x_3}) = \begin{vmatrix} e^{-t} & e^t & e^{3t} \\ 2e^{-t} & 0 & -e^{3t} \\ e^{-t} & e^t & 2e^{3t} \end{vmatrix} = -2e^{3t} \neq 0.$$

Therefore, the given set of vectors is linearly independent and so it forms a fundamental solution set. A fundamental matrix is given by

$$\Phi(t) = \begin{bmatrix} e^{-t} & e^t & e^{3t} \\ 2e^{-t} & 0 & -e^{3t} \\ e^{-t} & e^t & 2e^{3t} \end{bmatrix}.$$

The general solution is given by

$$\bar{x}(t) = \Phi(t)\bar{c} = c_1 \begin{bmatrix} e^{-t} \\ 2e^{-t} \\ e^{-t} \end{bmatrix} + c_2 \begin{bmatrix} e^t \\ 0 \\ e^t \end{bmatrix} + c_3 \begin{bmatrix} e^{3t} \\ -e^{3t} \\ 2e^{3t} \end{bmatrix}.$$

4. Find the general solution of the following first-order equations:

$$\begin{cases} \frac{dx}{dt} = 4x - y, \\ \frac{dy}{dt} = 2x + y. \end{cases}$$

**Solution:**
Given

$$\begin{cases} \frac{dx}{dt} = 4x - y, \\ \frac{dy}{dt} = 2x + y. \end{cases}$$

The auxiliary equation is $m^2 - 5m + 6 = 0$ giving the roots as $m_1 = 2, m_2 = 3$. The two linearly independent solutions are

$$\left.\begin{array}{l} x = A_1 e^{2t} \\ x = B_1 e^{2t} \end{array}\right\}, \quad \left.\begin{array}{l} x = A_2 e^{3t} \\ x = B_2 e^{3t} \end{array}\right\}.$$

Using the relation $(a_1 - m)A + b_1 B = 0$, we get $A_1 = 1, B_1 = 2, A_2 = 1, B_2 = 1$. Hence, the general solution can be written as

$$\left\{\begin{array}{l} x(t) = c_1 e^{2t} + c_2 e^{3t}, \\ y(t) = 2c_1 e^{2t} + c_2 e^{3t}. \end{array}\right.$$

5. Find the general solution of the following first-order equations:

$$\left\{\begin{array}{l} \frac{dx}{dt} = x + 3y, \\ \frac{dy}{dt} = 3x + y. \end{array}\right.$$

Then, find the particular solution given that $x(0) = 5$ and $y(0) = 1$.
**Solution:**
Given

$$\left\{\begin{array}{l} \frac{dx}{dt} = x + 3y, \\ \frac{dy}{dt} = 3x + y. \end{array}\right.$$

The auxiliary equation is $m^2 - 2m - 8 = 0$ giving the roots as $m_1 = -2, m_2 = 4$. The two linearly independent solutions are

$$\left.\begin{array}{l} x = A_1 e^{-2t} \\ x = B_1 e^{-2t} \end{array}\right\}, \quad \left.\begin{array}{l} x = A_2 e^{4t} \\ x = B_2 e^{4t} \end{array}\right\}.$$

Using the relation $(a_1 - m)A + b_1 B = 0$, we get $A_1 = 1, B_1 = -1, A_2 = 1, B_2 = 1$. Hence, the general solution can be written as

$$\left\{\begin{array}{l} x(t) = c_1 e^{-2t} + c_2 e^{4t}, \\ y(t) = -c_1 e^{-2t} + c_2 e^{4t}. \end{array}\right.$$

using the initial conditions $x(0) = 5, y(0) = 1$, we get $c_1 = 2, c_2 = 3$.
Therefore, the specific solution is given by

$$\left\{\begin{array}{l} x(t) = 2e^{-2t} + 3e^{4t}, \\ y(t) = -2e^{-2t} + 3e^{4t}. \end{array}\right.$$

6. Solve the given equation $\bar{x}'(t) = A\bar{x}(t)$ for

(a) $A = \begin{bmatrix} 0 & -2 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$, (b) $A = \begin{bmatrix} -1 & 1 & -2 \\ 0 & -1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$.

**Solution:** The solution is given by

$$\bar{x} = \alpha e^{\lambda t} \bar{c}.$$

We have to find the values of $\lambda$ from the characteristic equation $\det(\lambda I - A) = 0$ and the column vectors $\bar{c}$ from $(\lambda I - A) = \bar{0}$.
(a) The characteristic equation can be found as

$$(2 + \lambda)(\lambda^2 - 2\lambda + 2) = 0$$

We get the characteristic values as $\lambda = -2, 1+\mathrm{i}, 1-\mathrm{i}$.
Let $\bar{u}$ be the column vector corresponding to $\lambda = -2$. This can be obtained as

$$\bar{u} = [0 \quad 0 \quad 1]^T.$$

Let $\bar{v}$ be the column vector corresponding to $\lambda = 1+\mathrm{i}$. This can be obtained as

$$\bar{v} = [1-\mathrm{i} \quad -1 \quad 0]^T.$$

Let $\bar{w}$ be the column vector corresponding to $\lambda = 1-\mathrm{i}$. This can be obtained as

$$\bar{w} = [1+\mathrm{i} \quad -1 \quad 0]^T.$$

Hence, the general solution can be written as

$$\bar{x} = \alpha_1 \bar{u} e^{-2t} + \alpha_2 \bar{v} e^{(1+\mathrm{i})t} + +\alpha_3 \bar{w} e^{(1-\mathrm{i})t}.$$

(b) The characteristic equation gives the characteristic values as $\lambda = -1$ (repeated) and $\lambda = 1$.
Let $\bar{u}$ be the column vector corresponding to $\lambda = -1$. This can be obtained as

$$\bar{u} = [1 \quad 0 \quad 0]^T.$$

Let $\bar{v}$ be another column vector corresponding to the same root $\lambda = -1$. This can be obtained as

$$\bar{v} = [t \quad 0 \quad 0]^T.$$

Let $\bar{w}$ be the column vector corresponding to $\lambda = 1$. This can be obtained as

$$\bar{w} = [0 \quad 2 \quad 1]^T.$$

Hence, the general solution can be written as

$$\bar{x} = \alpha_1 \bar{u} e^{-t} + \alpha_2 \bar{v} e^{-t} + +\alpha_3 \bar{w} e^{t}.$$

7. Show that the solutions for the system

$$\begin{cases} \frac{dx}{dt} = x + 2y, \\ \frac{dy}{dt} = 3x + 2y. \end{cases}$$

are given by $\begin{cases} x = 2e^{4t}, \\ y = 3e^{4t}, \end{cases}$ and $\begin{cases} x = e^{-t}, \\ y = -e^{-t}. \end{cases}$
Further, show that the solutions of the above system can also be obtained by
(i) differentiating the first equation with respect to $t$ and then eliminating $y$,
(ii) differentiating the second equation with respect to $t$ and then eliminating $x$.
**Solution:** For the first part, proceed exactly in the same manner as in Problems 4 and 5. Then,
(i) Differentiate the first equation of the system to get

$$\frac{d^2x}{dt^2} = \frac{dx}{dt} + 2\frac{dy}{dt}. \tag{1}$$

Further, subtracting the first equation of the system from the second:

$$\frac{dy}{dt} = \frac{dx}{dt} + 2x. \tag{2}$$

Eliminating $\frac{dy}{dt}$ between equations (1) and (2), we get the following second-order equation in $x$:

$$\frac{d^2x}{dt^2} - 3\frac{dx}{dt} - 4x = 0.$$

Its general solution is

$$x(t) = c_1e^{4t} + c_2e^{-t}.$$

(ii) Differentiate the second equation of the system to get

$$\frac{d^2y}{dt^2} = 3\frac{dx}{dt} + 2\frac{dy}{dt}. \tag{3}$$

Further, subtracting the second equation of the system from the first one multiplied by 3:

$$3\frac{dx}{dt} = \frac{dy}{dt} + 4y. \tag{4}$$

Eliminating $\frac{dx}{dt}$ between equations (3) and (4), we get the following second-order equation in $y$:

$$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} - 4y = 0.$$

Its general solution is

$$y(t) = c_1e^{4t} + c_2e^{-t}.$$

8. We know that, for a system of two first-order equations, the complex roots $a \pm ib$ of the auxiliary equations lead to the following solutions:

$$\left.\begin{array}{l} x = e^{at}(A_1\cos bt - A_2\sin bt), \\ y = e^{at}(B_1\cos bt - B_2\sin bt), \end{array}\right\}$$

and

$$\left.\begin{array}{l} x = e^{at}(A_1\sin bt + A_2\cos bt), \\ y = e^{at}(B_1\sin bt + B_2\cos bt). \end{array}\right\}$$

Find the Wronskian of these solutions. Check whether it is mandatory that $A_1B_2 - A_2B_1 \neq 0$.

**Solution:**

We have

$$\begin{aligned} W(x(t), y(t)) &= \begin{vmatrix} x_1(t) & x_2(t) \\ y_1(t) & y_2(t) \end{vmatrix} \\ &= \begin{vmatrix} e^{at}(A_1\cos bt - A_2\sin bt) & e^{at}(A_1\sin bt + A_2\cos bt) \\ e^{at}(B_1\cos bt - B_2\sin bt) & e^{at}(B_1\sin bt + B_2\cos bt) \end{vmatrix} \\ &= e^{2at}(A_1B_2 - A_2B_1). \end{aligned}$$

In order that the solutions are linearly independent, we must have $A_1B_2 - A_2B_1 \neq 0$.

**The End**